

Software Design and Engineering

IS/IT Design Section

Cathedral of All Saints – IT Modernization

Albany, New York

Team Nine

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1. Introduction

This document presents the software design and engineering analysis for the Cathedral of All Saints IT Modernization project. It defines the key use cases that the proposed system must support, provides a detailed specification of a selected use case using the short-form template, and includes UML diagrams that model the system's behavior, structure, and deployment architecture.

The Cathedral of All Saints currently operates with minimal technology infrastructure. The proposed system centers on a WordPress-based content management system (CMS) integrated with online giving, event ticketing, a donor CRM, and cloud-based digital asset management. This software design ensures that staff members with limited technical experience can manage content, events, and communications effectively, while providing a modern, accessible experience for the general public.

2. Use Cases

The following use cases describe the primary interactions between actors (staff, public users, donors) and the Cathedral's IT system. Each use case captures a distinct goal that the system must support.

2.1 Use Case 1: Access Documents for Marketing Campaign

Actor: Staff Member

A staff member needs to create a marketing campaign and wants to use the CMS to access relevant information in one centralized location. They will look up the average donation amount from the donor/CRM system, review past marketing materials stored in cloud storage, and analyze which outreach strategies have been most effective. The system should allow them to efficiently pull together this data without needing to search across multiple disconnected tools, so they can target their next campaign effectively and with minimal effort.

2.2 Use Case 2: View Upcoming Events

Actor: General Public User

A member of the public wants to use the website to see upcoming events at the Cathedral. They need to be able to quickly and intuitively navigate to the events section without confusion. Once there, the events must be listed in a clear, chronological, and up-to-date format so the user does not have to decode a disorganized list. The goal is for the user to find what they need within seconds of arriving on the page, reducing friction and encouraging attendance.

2.3 Use Case 3: Update Website Content for Seasonal Schedule

Actor: Staff Member

A staff member wants to update the website to reflect the Cathedral's different schedule for the Christmas season. This update needs to be completed quickly and without risk of breaking other parts of the site. The CMS must provide a straightforward, block-based editing interface that does not require computer science expertise, so that any member of the Cathedral's lean team can make the change confidently. The system should isolate content areas to prevent unintended side effects when one section is updated.

2.4 Use Case 4: Set Up Event Ticketing

Actor: Staff Member

A staff member wants to set up ticketing for an upcoming Cathedral event. They need the system to allow them to enter general event information (date, time, description, ticket price), and then have the ticketing module handle the sales process, payment collection, and attendee tracking automatically. The ticketing system must be integrated directly into the Cathedral's website so that visitors can purchase tickets without being redirected to an unfamiliar third-party platform, and staff can view a uniform record of who has purchased tickets.

3. Use Case Specification (Short Form)

The following use case is specified in detail using the short-form template format.

Field	Description
Use Case Name	Looking at Upcoming Events
Goal	To let users easily see what events are coming up at the Cathedral of All Saints
Summary	The general public (user) is able to look up upcoming events that Cathedral staff members create and manage through the CMS. The end goal is to provide a single, consistent source of event information for the Cathedral community.
Primary Actor	General Public User
Secondary Actor	Staff Member (creates/manages events via CMS)
Preconditions	The Cathedral website is live and accessible; at least one upcoming event has been published by a staff member through the CMS.
Trigger	User navigates to the Cathedral of All Saints website in their web browser.
Basic Flow	1. User loads the Cathedral of All Saints homepage in their browser. 2. The website renders quickly with a clear navigation bar including an "Events" link. 3. User clicks the "Events" link in the navigation menu. 4. The browser sends a request to the web server, which queries the CMS database for all events with a date on or after today. 5. The server returns the events page, displaying event listings sorted chronologically with title, date, time, location, and a brief description. 6. User browses the event listings and optionally clicks on a specific event to view full details and ticket purchasing options.
Alternative Flows	3a. If no events are currently scheduled, the page displays a friendly message: "No upcoming events at this time. Check back soon!" 6a. If the user wants to purchase a ticket, they are directed to the integrated ticketing module without leaving the website.
Post Conditions	The user has viewed the events listing and understands what upcoming events are available. If desired, the user has accessed event detail pages and/or initiated a ticket purchase.
Notes	Website refers to the Cathedral of All Saints homepage (WordPress-based CMS). Events should be filterable by category in future iterations.
Resources	HTML/CSS/JS files, WordPress CMS, MySQL database, Event Tickets plugin, web hosting (SiteGround/AWS), domain URL

4. Use Case Diagram

The use case diagram below shows the primary actors and their interactions with the Cathedral IT system. Four actors are identified: the General Public User, Donor, Staff Member, and System Admin. The diagram illustrates which use cases each actor participates in and the relationships between use cases.

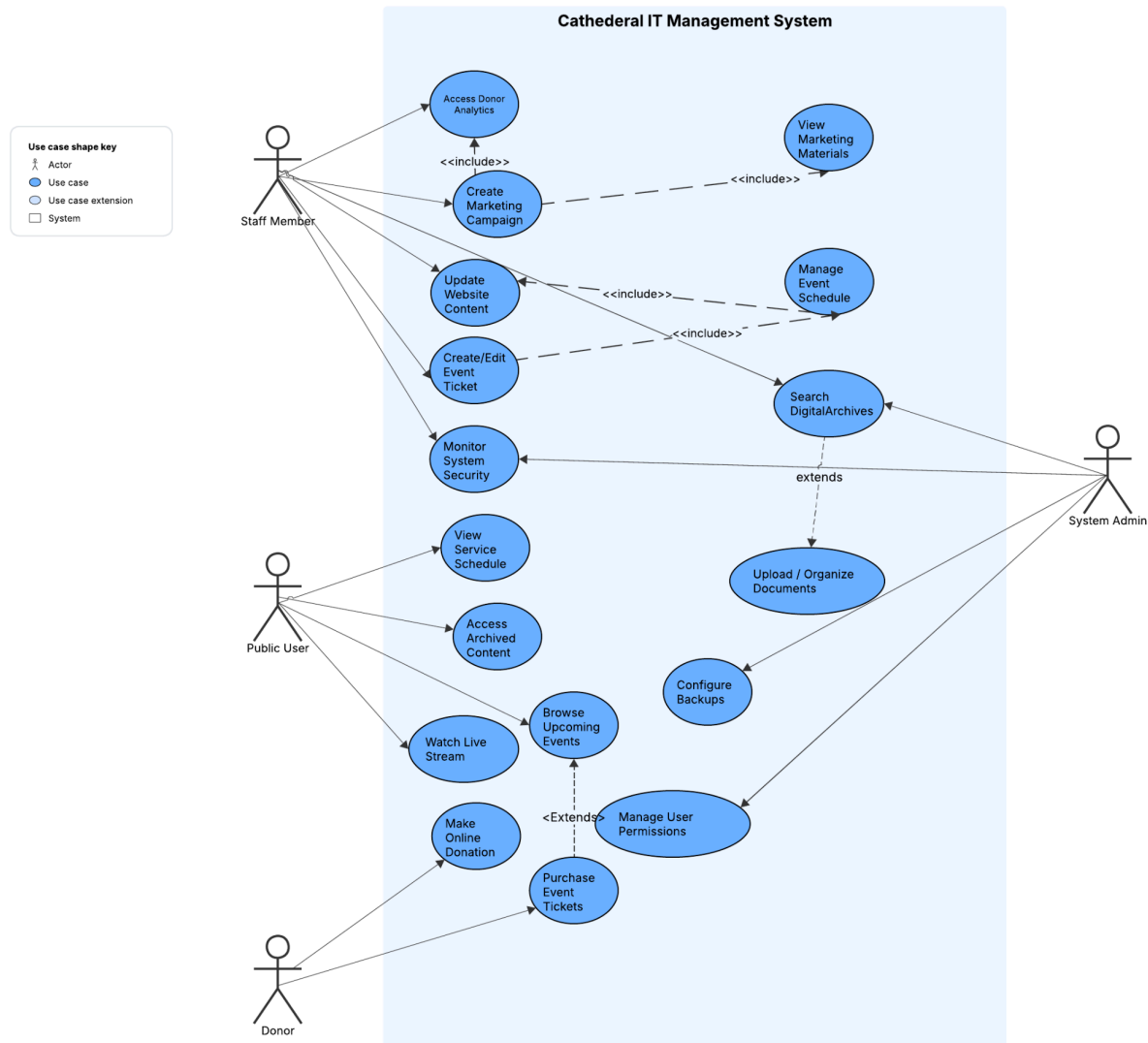


Figure 1: Use Case Diagram showing actors and their interactions with the Cathedral IT system. The General Public User and Donor interact with front-end use cases (viewing events, purchasing tickets, making donations), while Staff Members and System Admins interact with back-end management use cases (updating content, creating marketing campaigns). The «include» relationship indicates that purchasing tickets necessarily involves viewing events first.

Use Cases:

- A use case for the software in our project is a member of staff wanting to access documents to make a marketing campaign. For this case, they will want to use the CMS to go and easily look at the average donation amount in addition to what past marketing

material is all in one place so they can go and efficiently see the best way to target people in their next marketing campaign.

- b. Another use case is a member of the public wanting to use the website to see upcoming events going on at the Cathedral. For this case, the member will want to be able to quickly navigate to the event section in an easy to find way, so that they are not deterred. The events then need to be listed in an orderly and updated fashion so the user doesn't need to treat the list as a puzzle that they need to decode.
- c. A use case is a member of the staff wanting to update the site to put in their different schedule for the Christmas season. When they do this, they will want this to be done quickly and in a way that allows them to not have to worry about other parts of the site breaking when they update this info. They also would want this to be done in a way that doesn't require lots of computer science experience so that they can have any member of their lean team be able to update it.
- d. Another use case for a member of the staff is when they want to use it so that they can set up tickets for an event. In this case, they will want it to allow them to write the general info for the tickets, with the system then dealing with how the tickets are sold and allowing for the staff to see who got the tickets in a uniform way. They also will want the ticketing to be easily accessible and tied to their website so that they don't have to worry about having to explain a separate system to customers and other members of the staff.

5. Activity Diagram

The activity diagram below models the workflow for the "View Upcoming Events" use case. The diagram traces the complete flow from the user navigating to the website through the event page loading and optional detail selection.

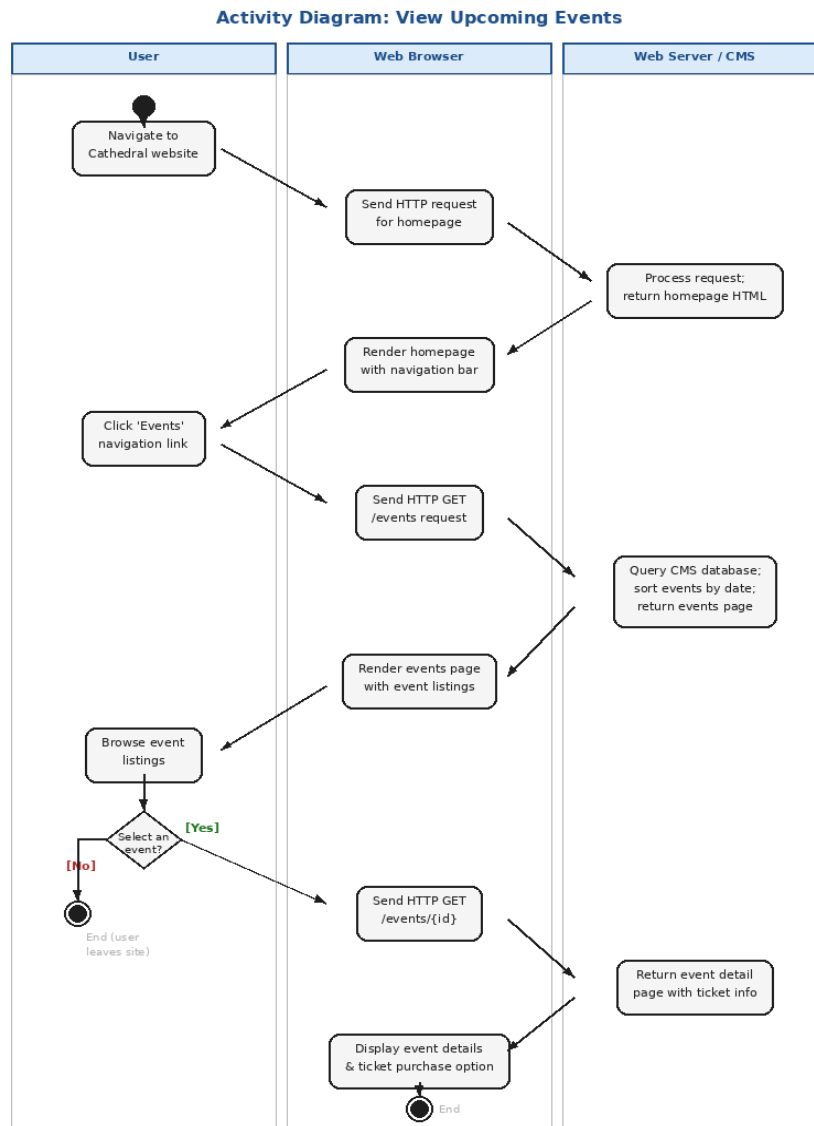


Figure 2:Activity Diagram for the “View Upcoming Events” use case. The diagram shows three swim lanes representing the User, Web Browser, and Server/CMS. The flow begins when the user navigates to the Cathedral website. The browser sends an HTTP request, the server processes it and returns the homepage. The user then clicks the Events link, triggering another request to the server, which queries the CMS database for active events sorted by date. The browser renders the event listings. A decision node allows the user to either browse and exit, or select a specific event to view its details and potentially purchase a ticket.

6. Sequence Diagram

The sequence diagram below illustrates the time-ordered interactions between the User, Web Browser, Web Server (WordPress), and CMS Database for the “View Upcoming Events” use case. Solid arrows represent requests, and dashed arrows represent responses.

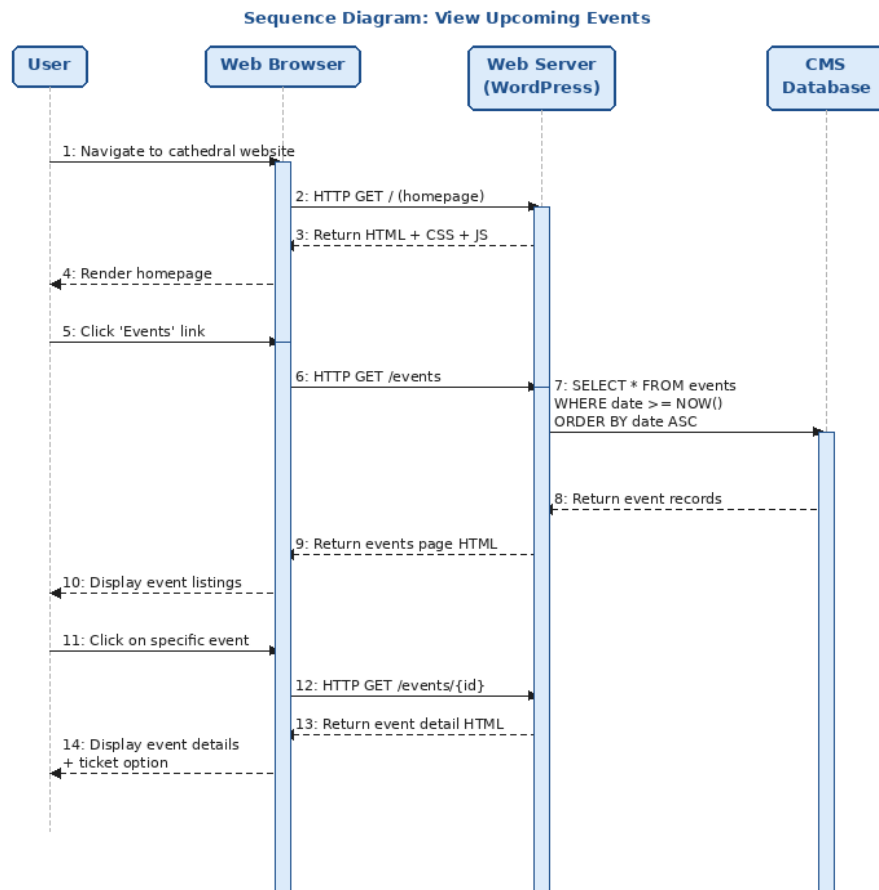


Figure 3: Sequence Diagram for the “View Upcoming Events” use case. The interaction begins when the User navigates to the Cathedral website (message 1). The Browser sends an HTTP GET request to the Web Server (2), which returns the homepage HTML (3). After the user clicks the Events link (5), the browser sends another GET request (6). The Web Server queries the CMS Database for events with a date on or after today, sorted in ascending order (7). The database returns event records (8), the server generates the events page (9), and the browser displays the listings (10). If the user clicks a specific event (11), a final request-response cycle retrieves and displays the event detail page with ticket purchasing options (12–14).

7. Component Diagram

The component diagram below shows the modular architecture of the Cathedral IT system. Each component is a self-contained, replaceable unit of the system. The diagram illustrates how front-end, back-end, and third-party service components interact through defined interfaces.

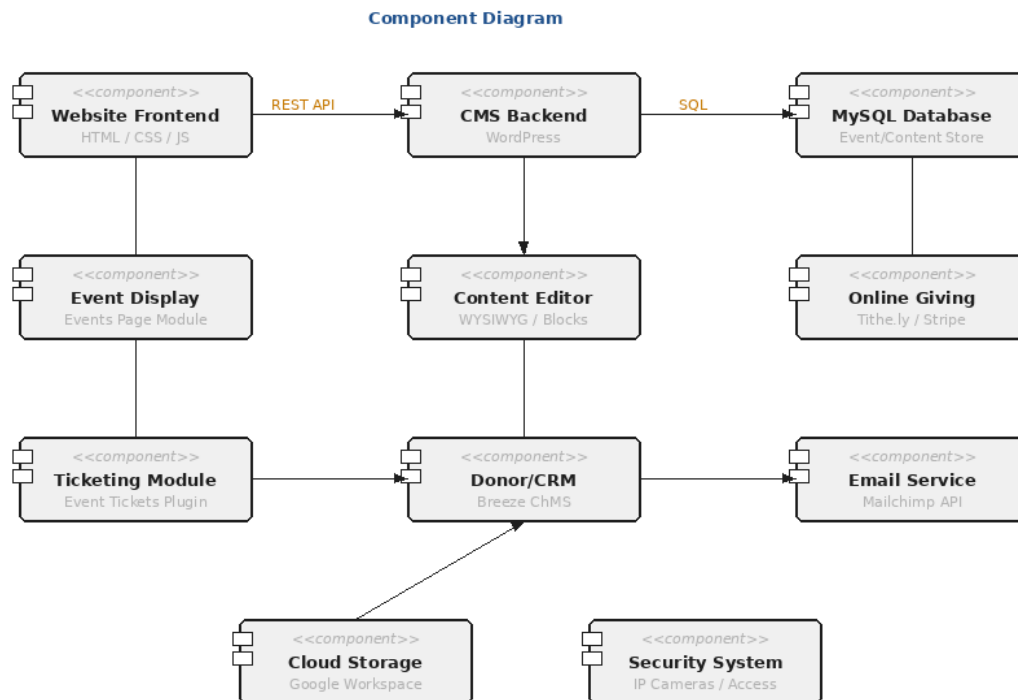


Figure 4: Component Diagram of the Cathedral IT system architecture. The Website Frontend (HTML/CSS/JS) communicates with the CMS Backend (WordPress) via REST API, which in turn connects to the MySQL Database via SQL. Supporting modules include the Event Display page, Content Editor (WYSIWYG block editor), Online Giving integration (Tithe.ly/Stripe), Ticketing Module (Event Tickets plugin), Donor/CRM system (Breeze ChMS), and Email Service (Mailchimp API). Cloud Storage (Google Workspace) provides digital asset management, and the Security System component manages IP cameras and access control. This modular design allows individual components to be upgraded or replaced without affecting the rest of the system.

8. Deployment Diagram

The deployment diagram below shows the physical architecture of the Cathedral IT system, including hardware nodes, the software artifacts deployed on each node, and the communication protocols connecting them.

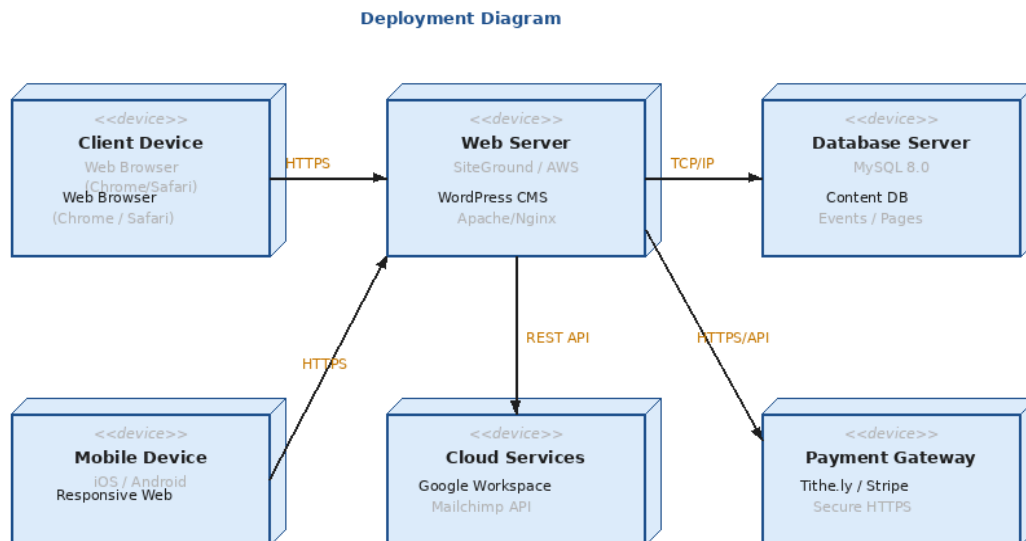


Figure 5: Deployment Diagram showing the physical infrastructure of the Cathedral IT system. Client devices (desktop and mobile browsers) connect to the Web Server via HTTPS. The Web Server runs WordPress CMS on Apache/Nginx and communicates with the MySQL Database Server over TCP/IP for content and event data storage. The Web Server also integrates with external Cloud Services (Google Workspace for storage, Mailchimp for email) via REST APIs, and with the Payment Gateway (Tithe.ly/Stripe) over secure HTTPS for online giving and ticket transactions. Mobile devices access the same system through the responsive web interface, ensuring a consistent experience across all device types.

9. Summary

This software design and engineering document has defined the key behavioral and structural models for the Cathedral of All Saints IT Modernization project. Four use cases were identified spanning the primary interactions of public users and staff members with the system. The “View Upcoming Events” use case was specified in detail using the short-form template, and five UML diagrams were produced:

- Use Case Diagram – Identifies actors and the system’s functional boundaries
- Activity Diagram – Models the step-by-step workflow with swim lanes showing responsibility allocation across user, browser, and server
- Sequence Diagram – Captures the time-ordered message exchange between system components
- Component Diagram – Defines the modular architecture and inter-component interfaces
- Deployment Diagram – Maps the physical infrastructure and communication protocols

Together, these models provide a comprehensive blueprint for the Cathedral’s IT system that supports implementation planning, stakeholder communication, and future system evolution. The modular, WordPress-based architecture ensures that staff with limited technical expertise can manage the system effectively, while the phased deployment strategy allows the Cathedral to adopt improvements incrementally.